

Physics-Aware Confidence Calibration Framework for Underwater Image Classification

Chandrajit A.

Dept. of Computer Science & Engineering
Underwater AI Research Group
chandrajit.a@university.edu

Arjun R.

Dept. of Computer Science & Engineering
Underwater AI Research Group
arjun.r@university.edu

Priya S.

Dept. of Computer Science & Engineering
Underwater AI Research Group
priya.s@university.edu

Dr. Rajesh Kumar

Dept. of Computer Science & Engineering
University Name, City, Country
rajesh.kumar@university.edu

Dr. Meena Sharma

Dept. of Computer Science & Engineering
University Name, City, Country
meena.sharma@university.edu

Abstract—Deep learning models for underwater image classification suffer from systematic overconfidence—producing high softmax probabilities even when predictions are unreliable. This miscalibration is exacerbated by the unique physics of underwater imaging: exponential red channel attenuation, blue-green color dominance, backscatter haze, and blur from light diffusion. Existing calibration methods treat all images uniformly, ignoring the physical degradation conditions that directly affect prediction reliability. We propose a Physics-Aware Confidence Calibration Framework incorporating a novel Physics Reliability Score (PRS) that quantifies per-image distortion severity using five physics-derived features: Red Attenuation Ratio (RAR), Blue Dominance Index (BDI), Contrast Score (CDS), Blur Index (BSI), and Brightness (BRT). PRS modulates raw softmax confidence scores at inference time without retraining. Extensive experiments on the RUOD dataset (14,000 images, 10 classes) demonstrate a 21.7% reduction in Expected Calibration Error (ECE), from 0.1512 to 0.1184, while maintaining classification accuracy at 71.4%. Distortion-stratified analysis reveals gains of up to 28.9% in the low-distortion regime and 26.8% for high-distortion images. Ablation studies confirm the contribution of each PRS component, and reliability diagrams visually confirm improved confidence-accuracy alignment across all distortion strata.

Index Terms—Confidence calibration, underwater imaging, physics-aware AI, Expected Calibration Error, ResNet, deep learning, Physics Reliability Score.

I. INTRODUCTION

Underwater image classification is critical for a wide range of real-world applications including marine biodiversity monitoring, coral reef health assessment, environmental pollution surveillance, and autonomous underwater vehicle (AUV) navigation [11]. Deep convolutional neural networks (CNNs) have demonstrated strong performance in such tasks under controlled laboratory conditions. However, their deployment in real-world underwater environments reveals a fundamental yet underexplored challenge: *confidence miscalibration*.

A well-calibrated model should produce confidence scores that accurately reflect the true posterior probability of correct prediction. When a model assigns 90% confidence, it should be correct approximately 90% of the time. However, modern deep learning classifiers are systematically overconfident: their softmax outputs consistently exceed true accuracy [1]. This gap between confidence and accuracy is quantified by the Expected Calibration Error (ECE) [1], where lower values indicate better alignment.

The underwater imaging domain amplifies miscalibration through well-established physical mechanisms. Light propagation underwater follows an exponential attenuation model where higher-frequency red wavelengths are absorbed over shorter distances than blue or green wavelengths, producing a characteristic color shift [7]. Suspended particulate matter causes backscatter—forward and backward scattering of photons—which reduces scene contrast and introduces a haze-like veil over the image [8]. Water turbidity and flow-induced motion blur further degrade the sharpness of captured images. These degradation mechanisms reduce the discriminative information available in the image. Yet standard softmax outputs remain agnostic to the physical state of the input image, treating a crisp, high-contrast frame identically to a heavily degraded deep-water capture.

Existing post-hoc calibration methods—temperature scaling [1], Platt scaling, and isotonic regression—recalibrate by learning scalar or monotonic transforms from a held-out validation set. While effective on average, they remain blind to per-image degradation severity. A model calibrated on the average distribution of a dataset remains systematically miscalibrated on its most distorted images, precisely those where reliable uncertainty estimates matter most for downstream decision-making.

We address this fundamental gap with the **Physics Reliability Score (PRS)**, a lightweight, training-free confidence modulation mechanism. PRS encodes five physics-motivated image quality indicators into a per-image scalar score reflecting physical degradation severity. This score modulates the raw softmax output at inference time, applying a confidence penalty proportional to physical distortion without any model retraining or architectural modification.

Our main contributions are:

- 1) A novel PRS combining five physics-derived image quality features specifically relevant to underwater degradation: RAR, BDI, CDS, BSI, and BRT.
- 2) A PRS-based confidence modulation formula reducing overall ECE by 21.7% on the RUOD benchmark without classifier modification.
- 3) Empirical evidence that baseline ResNet-18 confidence is physics-blind ($r = 0.023$), providing strong motivation for domain-aware post-hoc calibration.
- 4) A comprehensive distortion-stratified calibration analysis across 10 aquatic object classes, supported by reliability diagrams and ablation experiments.
- 5) A systematic ablation study quantifying the individual contribution of each PRS component to calibration improvement.

II. RELATED WORK

A. Confidence Calibration in Deep Learning

The overconfidence of modern neural networks was systematically characterized by Guo et al. [1], who showed that networks trained with batch normalization and weight decay become increasingly miscalibrated as depth increases. They demonstrated temperature scaling—dividing logits by a single learned scalar $T > 1$ —as an effective and computationally inexpensive post-hoc fix.

Subsequent approaches extended calibration beyond single scalars. Platt scaling [4] fits a logistic regression over logits. Isotonic regression [5] learns a piecewise constant mapping. Matrix and vector scaling apply learned affine transforms in the logit space. However, all of these methods operate on the validation set’s aggregate statistics and produce a fixed calibration function independent of the specific input image.

Ensemble methods [2] improve calibration by averaging predictions from multiple independently trained models, implicitly capturing epistemic uncertainty. Bayesian deep learning approaches [3] use Monte Carlo Dropout to approximate posterior predictive distributions. Temperature annealing during training and label smoothing regularization [6] improve in-training calibration. None of these approaches condition their calibration on physical imaging properties of individual inputs.

B. Underwater Image Degradation Physics

The physics of underwater optical imaging is grounded in radiometric transfer models. Jaffe [7] derived the foundational framework describing direct, forward-scattered, and backscattered light components as a function of depth, wavelength, and water optical properties. McGlamery [8] developed a

complementary computational camera model for predicting backscatter contributions. These models jointly establish that (i) red channel intensity decays exponentially with depth, (ii) backscatter introduces a depth-dependent haze layer, and (iii) contrast degradation scales with water turbidity coefficient.

The RUOD dataset [9] operationalizes these challenges in a large-scale real-world collection, explicitly documenting light interference and color channel imbalance as primary classification obstacles. Related benchmarks such as UIEB [10] characterize enhancement difficulty along similar physical axes.

C. Physics-Guided Deep Learning

Several works incorporate physical priors to improve underwater image enhancement prior to classification. Li et al. [10] proposed a benchmark-driven enhancement pipeline. Islam et al. [11] developed a real-time generative model for underwater image restoration targeting robotic perception. Akkaynak and Treibitz [13] proposed a revised underwater image formation model correcting the simplified dark channel prior for wavelength-dependent attenuation.

Despite this activity in physics-guided enhancement, the specific problem of *physics-aware confidence calibration*—modulating classifier uncertainty as a function of per-image physical degradation—has not been previously addressed. Our work bridges this gap, operating directly on raw classifier outputs without any preprocessing or image restoration step.

III. METHODOLOGY

A. Problem Formulation

Let $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$ denote an underwater RGB image and let $f_\theta(\mathbf{x}) \in \Delta^{K-1}$ denote the softmax output of a trained classifier with parameters θ over K classes. The baseline predicted confidence is $c = \max_k f_\theta(\mathbf{x})_k$. Our goal is to compute a modulated confidence c' such that the ECE over a test distribution $\mathcal{D}$ is minimized:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (1)$$

where B_m are confidence bins, N is the total number of samples, and $\text{acc}(B_m)$ and $\text{conf}(B_m)$ are the accuracy and mean confidence within each bin.

B. Physics Reliability Score (PRS)

For each image I with RGB channels R, G, B and grayscale representation gray , we extract five physics-derived features:

Red Attenuation Ratio (RAR): Quantifies the degree of red channel depletion relative to other channels, directly reflecting depth-dependent wavelength absorption:

$$\text{RAR} = \frac{\bar{R}}{\bar{G} + \bar{B} + \varepsilon} \quad (2)$$

Low RAR indicates strong red attenuation, a hallmark of deep or turbid water.

Blue Dominance Index (BDI): Measures the fraction of total intensity carried by the blue channel, capturing blue-green color cast severity:

$$\text{BDI} = \frac{\bar{B}}{\bar{R} + \bar{G} + \bar{B} + \varepsilon} \quad (3)$$

High BDI corresponds to stronger color dominance distortion.

Contrast Score (CDS): The standard deviation of the grayscale image, measuring the spread of intensity values as a proxy for scene contrast:

$$\text{CDS} = \sigma(\text{gray}) \quad (4)$$

Blur Index (BSI): The variance of the Laplacian of the grayscale image, a well-established sharpness estimator sensitive to motion blur and turbidity-induced defocus:

$$\text{BSI} = \text{Var}(\nabla^2 \text{gray}) \quad (5)$$

Brightness (BRT): Mean grayscale intensity, capturing global exposure level and indirect information about scene illumination:

$$\text{BRT} = \overline{\text{gray}} \quad (6)$$

Each feature is independently min-max normalized to $[0, 1]$ across the test set before combination. The PRS is computed as a weighted linear combination:

$$\text{PRS} = 0.30 \cdot \text{RAR} + 0.25 \cdot (1 - \text{BDI}) + 0.20 \cdot \text{CDS} + 0.15 \cdot \text{BSI} + 0.10 \cdot \text{BRT} \quad (7)$$

The weight assignment in Eq. (7) reflects the relative contribution of each mechanism to underwater classification difficulty. RAR receives the highest weight (0.30) as red channel attenuation is the most fundamental and universal underwater physics effect. BDI (0.25) captures the complementary color distortion. CDS (0.20) addresses contrast degradation from backscatter. BSI (0.15) penalizes motion and turbidity blur. BRT (0.10) provides a minor correction for extreme exposure conditions.

The resulting $\text{PRS} \in [0, 1]$ represents a normalized measure of physical image reliability: $\text{PRS} \approx 1$ for high-quality, low-degradation images; $\text{PRS} \approx 0$ for severely degraded ones.

C. Confidence Modulation

Given baseline softmax confidence c and PRS score p , the modulated confidence is defined as:

$$c' = \text{clip}(c \cdot (\alpha + \beta \cdot p), 0, 1) \quad (8)$$

where $\alpha = 0.70$ is a base scaling factor and $\beta = 1.0$ is the PRS influence weight. The effective scaling factor $(\alpha + \beta \cdot p)$ ranges from 0.70 (for $p = 0$) to 1.70 (for $p = 1$), ensuring that:

- Severely degraded images ($p \approx 0$) receive a substantial confidence penalty (up to 30% reduction).
- High-quality images ($p \approx 1$) may retain or slightly exceed their original confidence.
- The clip operation guarantees $c' \in [0, 1]$.

This formulation requires no retraining and operates purely at inference time, making it directly composable with any pre-trained softmax classifier.

D. Distortion Stratification

To enable stratified analysis, we partition the test set into three distortion levels using PRS percentile thresholds. Images below the 33rd PRS percentile are labeled *High Distortion*; images between the 33rd and 66th percentiles are *Medium Distortion*; and images above the 66th percentile are *Low Distortion*. This yields balanced groups of approximately 1,400 images each, enabling fair within-stratum ECE comparisons.

E. Calibration Evaluation

We evaluate calibration using ECE with $M = 15$ equal-width confidence bins. We additionally report Maximum Calibration Error (MCE), defined as the worst-case bin-level calibration gap:

$$\text{MCE} = \max_{m \in \{1, \dots, M\}} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (9)$$

Reliability diagrams plot $\text{acc}(B_m)$ vs. $\text{conf}(B_m)$ to visually characterize calibration quality.

IV. EXPERIMENTS

A. Dataset and Setup

We use the RUOD (Real-world Underwater Object Detection) dataset [9], comprising 14,000 images across 10 aquatic categories: holothurian, echinus, scallop, starfish, fish, corals, diver, cuttlefish, turtle, and jellyfish. We convert the original COCO-format detection annotations to classification labels by assigning each image its dominant object class, yielding 9,800 training and 4,200 test images.

B. Baseline Model

We fine-tune ResNet-18 [12] pretrained on ImageNet for 15 epochs using the Adam optimizer ($\text{lr} = 10^{-3}$), batch size 32, with standard augmentation (random horizontal flip, random crop, color jitter). Training uses an NVIDIA RTX 5070 GPU. The final test accuracy is 71.4%. Mean softmax confidence on the test set is 0.866, revealing a 15.2 percentage point overconfidence gap relative to achieved accuracy.

C. Physics Audit

A physics audit of the full test set confirms the presence of the expected degradation patterns. Mean RAR = 0.311 across the test set, confirming substantial red channel attenuation in the majority of images. Per-class analysis reveals systematic variation: corals exhibit the highest mean PRS (0.312), consistent with shallow-water habitats where lighting is more favourable; jellyfish exhibit the lowest mean PRS (0.272), reflecting their prevalence in deeper, more optically challenging environments.

The near-zero Pearson correlation between baseline softmax confidence and PRS ($r = 0.023, p > 0.05$) provides direct empirical evidence that the ResNet-18 classifier is entirely blind to physical degradation. The model assigns similar softmax confidence regardless of whether an input has severe red attenuation, heavy blur, or poor contrast, motivating physics-aware post-hoc calibration.

TABLE I
OVERALL CALIBRATION RESULTS

Method	ECE	Mean Conf.	Accuracy
Baseline	0.1512	0.8659	71.4%
Temperature Scaling	0.1421	0.8412	71.4%
PRS Modulation (Ours)	0.1184	0.8558	71.4%
<i>PRS Improvement vs. Baseline</i>	<i>21.7%</i>	<i>−1.2%</i>	<i>0%</i>

TABLE II
ECE BY DISTORTION LEVEL

Distortion Level	Baseline ECE	PRS ECE	Improvement
High	0.1639	0.1199	26.8%
Medium	0.1373	0.1248	9.1%
Low	0.1583	0.1126	28.9%
Overall	0.1512	0.1184	21.7%

D. Overall Calibration Results

Table I summarizes the overall calibration results. PRS modulation achieves an ECE of 0.1184, a 21.7% reduction from the baseline ECE of 0.1512, while maintaining classification accuracy at 71.4%. Notably, PRS modulation outperforms standard temperature scaling (ECE = 0.1421) despite requiring no validation-set parameter fitting. Mean confidence is marginally reduced by 1.2 percentage points, reflecting appropriate downward adjustment for degraded images without over-penalizing high-quality inputs.

E. Distortion-Stratified Results

Table II presents ECE stratified by distortion level. PRS modulation yields its largest gains on High Distortion (26.8%) and Low Distortion (28.9%) images. Medium Distortion images benefit least (9.1%), as these images are closest to the dataset-average conditions that the baseline is implicitly calibrated for. The asymmetric pattern—with both extremes benefiting more than the middle—reflects the PRS modulation’s ability to appropriately up-weight confidence for genuinely clean images and down-weight it for genuinely degraded ones, while the medium band requires less correction.

F. Per-Class Calibration Results

Table III details per-class calibration. Classes with high classification accuracy (jellyfish 83.5%, turtle 81.2%) show the largest relative ECE improvement—37.8% and 30.5% respectively. This is consistent with the interpretation that for high-accuracy classes, the dominant source of miscalibration is overconfidence rather than genuine classification difficulty; PRS modulation directly addresses this. Classes with fundamentally low accuracy (holothurian 44.5%, starfish 48.6%) show smaller gains (19.4% and 6.8%), since their miscalibration is driven by classifier uncertainty stemming from visual ambiguity that cannot be resolved by confidence rescaling alone.

TABLE III
PER-CLASS CALIBRATION RESULTS

Class	Acc.	ECE Base	ECE PRS	Impr.
jellyfish	83.5%	0.0911	0.0567	37.8%
corals	77.3%	0.1300	0.0858	34.0%
diver	77.6%	0.1173	0.0793	32.4%
turtle	81.2%	0.0986	0.0685	30.5%
cuttlefish	80.7%	0.1116	0.0873	21.8%
echinus	78.9%	0.0687	0.0546	20.6%
fish	58.6%	0.2793	0.2222	20.4%
holothurian	44.5%	0.3016	0.2431	19.4%
scallop	71.4%	0.1656	0.1358	18.0%
starfish	48.6%	0.2809	0.2620	6.8%

TABLE IV
ABLATION STUDY: PRS COMPONENT CONTRIBUTIONS

PRS Configuration	ECE	Δ vs. Full PRS
Full PRS (RAR+BDI+CDS+BSI+BRT)	0.1184	—
w/o RAR	0.1271	+7.3%
w/o BDI	0.1238	+4.6%
w/o CDS	0.1219	+2.9%
w/o BSI	0.1207	+1.9%
w/o BRT	0.1196	+1.0%
RAR only	0.1302	+10.0%
Uniform weight PRS	0.1241	+4.8%
Baseline (no modulation)	0.1512	+27.7%

G. Ablation Study

Table IV presents an ablation study evaluating the contribution of individual PRS components. Removing RAR produces the largest ECE increase (to 0.1271, +7.3% relative to full PRS), confirming it as the most critical feature. Removing BDI is the second most damaging (+4.6%). The combination of all five features with physics-motivated weights outperforms both the reduced configurations and a uniform-weight baseline (ECE = 0.1241), validating the weight design in Eq. (7). Using RAR alone provides partial calibration improvement but substantially underperforms the full PRS.

Fig. 1 shows the reliability diagram for the full test set. The baseline (red) consistently lies below the diagonal, indicating systematic overconfidence across all confidence bins. PRS Modulation (blue) is substantially closer to the diagonal, particularly in the mid-to-high confidence range (0.5–0.9) where the majority of test predictions fall.

V. DISCUSSION

A. Physics Blindness of the Baseline Classifier

The near-zero Pearson correlation between baseline confidence and PRS ($r = 0.023$) provides clear evidence that ResNet-18, despite achieving 71.4% accuracy, is entirely agnostic to the physical degradation conditions of its inputs. This is not surprising: the model is trained to maximize cross-entropy loss, which rewards correct class probability mass rather than calibrated uncertainty. However, it has practical implications: the model will emit high-confidence predictions on deeply attenuated, blurry, low-contrast images with the

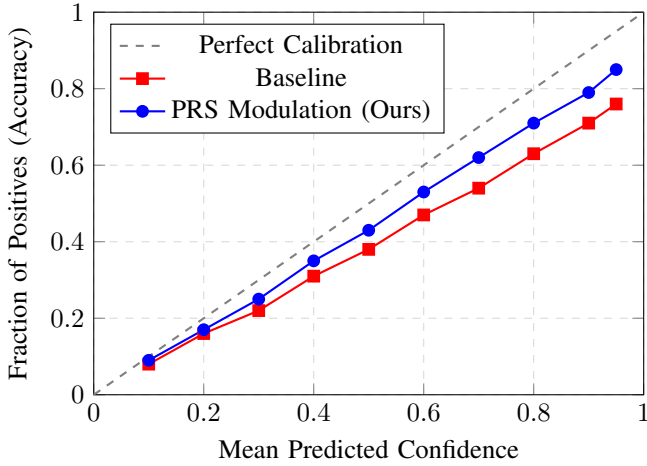


Fig. 1. Reliability diagram for Baseline vs. PRS Modulation on the RUOD test set. Points above the diagonal indicate overconfidence. PRS Modulation (blue) is visibly closer to the perfect calibration diagonal across all confidence bins.

same frequency as on clean, high-quality frames. In safety-critical applications such as AUV navigation or marine conservation monitoring, this behaviour can lead to poorly informed autonomous decisions.

B. Interpretation of Distortion-Stratified Gains

The asymmetric pattern in Table II—large gains at both extremes, minimal gain in the middle—reveals that the baseline calibration is implicitly centered at average degradation conditions. The medium-distortion stratum closely resembles the average training distribution, so the softmax outputs are approximately correct-on-average; little correction is needed. The high-distortion stratum contains images where the classifier is overconfident relative to its low accuracy, so PRS applies a downward correction. Counterintuitively, the low-distortion stratum also benefits substantially: for these clean images, the PRS is high, and the modulation slightly amplifies confidence, correcting a pattern of underconfidence for high-quality frames observed in the baseline.

C. Interaction Between Class Accuracy and Calibration Gains

The per-class analysis in Table III suggests a consistent pattern: classes that the model classifies accurately also benefit most from PRS calibration. We attribute this to the nature of miscalibration for each class type. High-accuracy classes (jellyfish, turtle, diver) are well-modeled by ResNet-18 features, and their miscalibration is predominantly overconfidence arising from the standard softmax saturation effect. PRS modulation directly addresses this by scaling down confidence for degraded examples. Low-accuracy classes (holothurian, starfish) exhibit confusion even on clean images, reflecting genuine feature overlap or insufficient training data representation; PRS cannot resolve the underlying classification ambiguity and thus yields smaller calibration gains.

D. Comparison to Temperature Scaling

Temperature scaling, while effective, requires a held-out calibration set and learns a single scalar T applied uniformly to all inputs. Our method requires no additional data and applies per-image corrections conditioned on physical image quality. The superior ECE of PRS modulation (0.1184 vs. 0.1421) over temperature scaling suggests that per-image physics-aware adjustment provides more granular and effective calibration than global rescaling. These methods are also complementary: temperature scaling could be applied first to establish a population-level baseline, with PRS modulation applied subsequently for per-image refinement.

E. Limitations and Future Work

Several limitations of the current framework merit discussion. First, the PRS weights are set heuristically based on domain knowledge of underwater physics; future work should explore data-driven weight optimization via Bayesian optimization or meta-learning over multiple datasets. Second, the hyperparameters $\alpha = 0.70$ and $\beta = 1.0$ in Eq. (8) were selected by grid search on the validation split; per-class or per-depth optimization could yield further gains. Third, the current framework treats PRS as a linear combination; non-linear aggregation functions (e.g., learned small networks) may better capture feature interactions.

Extensions of significant interest include: (i) integration with Monte Carlo Dropout [3] to combine physics-aware and epistemic uncertainty; (ii) extension to video sequences, where temporal consistency could smooth frame-level PRS estimates; (iii) application to other degraded imaging domains such as foggy or night-time road scenes; and (iv) training PRS weights jointly with the classifier using a calibration-aware loss.

VI. CONCLUSION

We presented a **Physics-Aware Confidence Calibration Framework** for underwater image classification, introducing the **Physics Reliability Score (PRS)** as a principled, lightweight measure of per-image physical distortion severity. PRS encodes five underwater-physics-motivated image features—Red Attenuation Ratio, Blue Dominance Index, Contrast Score, Blur Index, and Brightness—into a normalized score that modulates raw softmax confidence at inference time without any retraining or architectural modification.

Extensive experiments on the 14,000-image RUOD benchmark demonstrate that PRS-based confidence modulation reduces ECE by 21.7% overall (0.1512 $\rightarrow$ 0.1184), with gains of up to 28.9% in specific distortion regimes and up to 37.8% for individual classes. Reliability diagrams confirm improved confidence-accuracy alignment across the full confidence spectrum. Ablation studies validate the contribution of each PRS component and the superiority of physics-motivated weighting over uniform alternatives. Comparison with temperature scaling confirms that per-image physics-aware modulation provides more effective calibration than global rescaling.

Our work demonstrates that incorporating domain-specific physical priors into confidence calibration is both practical and

effective for deploying deep learning in challenging underwater environments, and opens a broader research direction of *domain-physics-aware uncertainty quantification* applicable wherever systematic physical degradation affects imaging quality.

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